Formal Truth and Modal Logic

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University of BRISTOL

Introduction

Modal Logic and Sentential Predicates

- Modal logics as logics of sentential predicates;
  - Provability logic as logic of the provability predicate;
  - Modal logic [truth logic] as logic of the truth predicate?

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Truth logics?

Are there candidate modal logics that are logics of the (a) truth predicate?

A primer on formal truth

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- Truth can be axiomatized!
  - Replace convention T by consistent sets of truth principles;
  - Approximating Convention T?
  - "Axiomatic truth theories"

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Remark

Truth theories require some syntax theory, e.g., arithmetics.

Logics of Truth?

What is the logic of truth of a given theory of truth?

- What are the unrestricted rules and principles of truth?
- What is the modal logic of a truth theory?
- Which modal logic captures the logical properties of the truth predicate of a given truth theory?

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Learning from Provability

Similar questions have been explored for the provabilily predicate:

- Solovay showed that GL was the logic of the provability predicate of PA.

Provability: Arithmetical Interpretation

Syntax of $\mathcal{L}_{\Box}$:

$$\varphi ::= p \mid \bot \mid \top \mid \neg\varphi \mid \varphi \wedge \varphi \mid \Box\varphi$$

where $p \in \mathrm{At}$.

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A function $* : \mathrm{At} \to \mathrm{Sent}_{\mathcal{L}_{\mathrm{PA}}}$ is called a realization.

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Arithmetical Interpretation

A realization $*$ induces an arithmetical interpretation $H^* : \mathcal{L}_\Box \to \mathrm{Sent}_{\mathcal{L}_{\mathrm{PA}}}$:

$$H^*(\varphi) := \begin{cases} \varphi^*, & \text{if } \varphi \in \mathrm{At}; \\ 0 = 1, & \text{if } \varphi \doteq \bot; \\ 0 = 0, & \text{if } \varphi \doteq \top; \\ \neg H^*(\psi), & \text{if } \varphi \doteq \neg\psi; \\ H^*(\psi) \wedge H^*(\chi), & \text{if } \varphi \doteq \psi \wedge \chi; \\ \mathrm{Pr}_{\mathrm{PA}}(\ulcorner H^*(\psi) \urcorner), & \text{if } \varphi \doteq \Box\psi. \end{cases}$$

Provability: Solovay's Completeness Result

Theorem (Solovay)

For all $\varphi \in \mathcal{L}_\Box$

$$\mathrm{GL} \vdash \varphi \Leftrightarrow \forall * \,(\mathrm{PA} \vdash H^*(\varphi))$$
$$\mathrm{GLS} \vdash \varphi \Leftrightarrow \forall * \,(\mathbb{N} \models H^*(\varphi))$$

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- Axiomatic theories of truth: isn't it trivial?
  - Löb's theorem

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  - T-sentences

Truth Interpretation

Truth-Realization

A function $* : \mathrm{At} \to \mathrm{Sent}_{\mathcal{L}_{\mathrm{T}}}$ is called a truth-realization.

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A function $*$: At $\to$ Sent$_{\mathcal{L}_\mathrm{T}}$ is called a truth-realization.

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A realization $*$ induces a truth interpretation $H^* : \mathcal{L}_\Box \to$ Sent$_{\mathcal{L}_\mathrm{T}}$:

$$H^*(\varphi) := \begin{cases} \varphi^*, & \text{if } \varphi \in \mathrm{At}; \\ 0 = 1, & \text{if } \varphi \doteq \bot; \\ 0 = 0, & \text{if } \varphi \doteq \top; \\ \neg H^*(\psi), & \text{if } \varphi \doteq \neg\psi; \\ H^*(\psi) \wedge H^*(\chi), & \text{if } \varphi \doteq \psi \wedge \chi; \\ \mathrm{T}\ulcorner H^*(\psi) \urcorner, & \text{if } \varphi \doteq \Box\psi. \end{cases}$$

Towards Truth-theoretic Completeness

Let Σ be a truth theory. Then for all $\varphi \in \mathcal{L}_\Box$

$$? \vdash \varphi \text{ iff } \forall * (\Sigma \vdash H^*(\varphi)).$$

Aim of talk

Determine the modal logics of Kripkean truth theories;

- Kripke-Feferman Truth

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Determine the modal logics of Kripkean truth theories;

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- Truth in Kripkean fixed-point models

- The results are joint work with Carlo Nicolai (Nicolai and Stern, 2021).
- Philosophical discussion in Stern (2025)

Outline

Introduction

Kripkean Truth

Modal Logics and Semantics

Truth-theoretic Completeness

Refutation Proofs

Kripkean Truth

KF extends PAT with the axioms:

(KF1)	$\forall s \forall t(\mathrm{T}s \mathbin{\underset{\cdot}{=}} t \leftrightarrow s^{\circ} = t^{\circ})$
(KF2)	$\forall s \forall t(\mathrm{T}s \mathbin{\underset{\cdot}{\neq}} t \leftrightarrow s^{\circ} \neq t^{\circ})$
(KF3)	$\forall x(\mathrm{Sent}(x) \to (\mathrm{T}\underset{\cdot}{\neg}\underset{\cdot}{\neg}x \leftrightarrow \mathrm{T}x))$
(KF4)	$\forall x, y(\mathrm{Sent}(x \mathbin{\underset{\cdot}{\wedge}} y) \to (\mathrm{T}x \wedge \mathrm{T}y \leftrightarrow \mathrm{T}(x \mathbin{\underset{\cdot}{\wedge}} y)))$
(KF5)	$\forall x, y(\mathrm{Sent}(x \mathbin{\underset{\cdot}{\wedge}} y) \to (\mathrm{T}\underset{\cdot}{\neg}x \vee \mathrm{T}\underset{\cdot}{\neg}y \leftrightarrow \mathrm{T}\underset{\cdot}{\neg}(x \mathbin{\underset{\cdot}{\wedge}} y)))$
(KF6)	$\forall v \forall x(\mathrm{Sent}(\underset{\cdot}{\forall}vx) \to (\forall t \mathrm{T}x(t/v) \leftrightarrow \mathrm{T}\underset{\cdot}{\forall}vx))$
(KF7)	$\forall v \forall x(\mathrm{Sent}(\underset{\cdot}{\forall}vx) \to (\exists t \mathrm{T}\underset{\cdot}{\neg}x(t/v) \leftrightarrow \mathrm{T}\underset{\cdot}{\neg}\underset{\cdot}{\forall}vx))$
(KF8)	$\forall t(\mathrm{T}t^{\circ} \leftrightarrow \mathrm{T}\,\underset{\cdot}{T}\,t)$
(KF9)	$\forall t(\mathrm{T}\underset{\cdot}{\neg}t^{\circ} \vee \neg\mathrm{Sent}(t^{\circ}) \leftrightarrow \mathrm{T}\underset{\cdot}{\neg}\underset{\cdot}{\mathrm{T}}\,t)$

Axiomatic Truth: The Theory Kripke-Feferman

KF+CN

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(CN)	$\forall x (\mathrm{Sent}(\mathrm{x}) \rightarrow (\mathrm{T} \underset{\cdot}{\neg} x \rightarrow \neg \mathrm{T} x))$

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Kripke's semantic theory: Fixed-point Semantics

- Let e be a evaluation scheme. Then e gives rise to a jump operation $\mathcal{J}_{\mathrm{e}} : \mathcal{P}(\omega) \to \mathcal{P}(\omega)$ with

$$\mathcal{J}_{\mathrm{e}}(X) = \{\#\phi : (\mathbb{N}, X) \vDash_{\mathrm{e}} \phi\}$$

  for $X \subseteq \omega$.

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  - K3 Consistent fixed points—KF+CN
  - LP Complete fixed points—KF+CM

A Solovay-Style Result for Kripke-Feferman?

The Question

$$\mathbf{?} \vdash \varphi \Leftrightarrow \forall * \left(\mathsf{KF} \vdash H^*(\varphi)\right)$$

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An Idea

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- Inner logic is subclassical, i.e., FDE.

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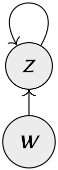

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- The classical world *w* sees an “idiosyncratic” subclassical world *z*.

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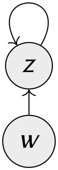

- Classical logic but..
- Inner logic is subclassical, i.e., FDE.
- The classical world w sees an "idiosyncratic" subclassical world z.
- The logic of w is the modal logic of KF.

Modal Logics and Semantics

Subclassical Logics

FDE

(REF) $\Gamma, \varphi \Rightarrow \varphi, \Delta$ for φ a literal

(CUT) $$\frac{\Gamma \Rightarrow \varphi, \Delta \qquad \Gamma, \varphi \Rightarrow \Delta}{\Gamma \Rightarrow \Delta}$$

($\bot$) $\Gamma, \bot \Rightarrow \Delta$

($\top$) $\Gamma \Rightarrow \top, \Delta$

(dn-l) $$\frac{\Gamma \Rightarrow \varphi, \Delta}{\Gamma \Rightarrow \neg\neg\varphi, \Delta}$$

(dn-r) $$\frac{\Gamma, \varphi \Rightarrow \Delta}{\Gamma, \neg\neg\varphi \Rightarrow \Delta}$$

($\wedge$l) $$\frac{\Gamma, \varphi, \psi \Rightarrow \Delta}{\Gamma, \varphi \wedge \psi \Rightarrow \Delta}$$

($\wedge$r) $$\frac{\Gamma \Rightarrow \varphi, \Delta \qquad \Gamma \Rightarrow \psi, \Delta}{\Gamma \Rightarrow \varphi \wedge \psi, \Delta}$$

($\neg\wedge$r) $$\frac{\Gamma, \neg\varphi \Rightarrow \Delta \qquad \Gamma, \neg\psi \Rightarrow \Delta}{\Gamma, \neg(\varphi \wedge \psi) \Rightarrow \Delta}$$

($\neg\wedge$r) $$\frac{\Gamma \Rightarrow \neg\varphi, \neg\psi, \Delta}{\Gamma \Rightarrow \neg(\varphi \wedge \psi), \Delta}$$

Subclassical Logics

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Subclassical Logics

LP

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The Truth Logic $\mathrm{FDE}_{\Box}$

The truth logic of the subclassical worlds

Let $\mathcal{S} \in \{\mathrm{FDE}, \mathrm{K3}, \mathrm{LP}\}$. Then $\mathcal{S}_{\Box}$ is obtained by adding the following rules to $\mathcal{S}$:

$$(\Box\mathrm{L}) \quad \frac{\Gamma, \varphi \Rightarrow \Delta}{\Gamma, \Box\varphi \Rightarrow \Delta} \qquad (\Box\mathrm{R}) \quad \frac{\Gamma \Rightarrow \varphi, \Delta}{\Gamma \Rightarrow \Box\varphi, \Delta}$$

$$(\neg\Box\mathrm{L}) \quad \frac{\Gamma, \neg\varphi \Rightarrow \Delta}{\Gamma, \neg\Box\varphi \Rightarrow \Delta} \qquad (\neg\Box\mathrm{R}) \quad \frac{\Gamma \Rightarrow \neg\varphi, \Delta}{\Gamma \Rightarrow \neg\Box\varphi, \Delta}$$

Non-Congruent Modal Logic

The logic of the classical worlds

BM extends propositional logic by

$(\top)$	$\Box\top$
$(\bot)$	$\neg\Box\bot$
$(\neg)$	$\Box\varphi \leftrightarrow \Box\neg\neg\varphi$
$(\wedge 1)$	$\Box(\varphi \wedge \psi) \leftrightarrow \Box\varphi \wedge \Box\psi$
$(\wedge 2)$	$\Box\neg(\varphi \wedge \psi) \leftrightarrow \Box\neg\varphi \vee \Box\neg\psi$
$(\Box 1)$	$\Box\varphi \leftrightarrow \Box\Box\varphi$
$(\Box 2)$	$\Box\neg\varphi \leftrightarrow \Box\neg\Box\varphi$
(FV)	$\Box\varphi \wedge \neg\Box\neg\varphi \rightarrow \varphi$

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$(\Box 1)$	$\Box\varphi \leftrightarrow \Box\Box\varphi$
$(\Box 2)$	$\Box\neg\varphi \leftrightarrow \Box\neg\Box\varphi$
(FV)	$\Box\varphi \wedge \neg\Box\neg\varphi \rightarrow \varphi$
(D)	$\Box\neg\varphi \rightarrow \neg\Box\varphi$

Non-Congruent Modal Logic

The logic of the classical worlds

M^b extends propositional logic by

$(\top)$	$\Box\top$
$(\bot)$	$\neg\Box\bot$
$(\neg)$	$\Box\varphi \leftrightarrow \Box\neg\neg\varphi$
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$(\Box 1)$	$\Box\varphi \leftrightarrow \Box\Box\varphi$
$(\Box 2)$	$\Box\neg\varphi \leftrightarrow \Box\neg\Box\varphi$
(FV)	$\Box\varphi \wedge \neg\Box\neg\varphi \to \varphi$
(D_c)	$\neg\Box\varphi \to \Box\neg\varphi$

The Connection

- The subclassical truth logic is the **inner** logic of the non-congruent modal logic.

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Connecting Lemma

Let Γ, Δ be finite sets of formulas of $\mathcal{L}_{\Box}$. Then

$$\text{If } \mathsf{FDE}_{\Box} \vdash \Gamma \Rightarrow \Delta, \text{ then } \mathsf{BM} \vdash \bigwedge \Box\Gamma \to \bigvee \Box\Delta$$

This also holds for the pairs $(\mathsf{K3}_{\Box}, \mathsf{M}^{\mathsf{n}})$ and $(\mathsf{LP}_{\Box}, \mathsf{M}^{\mathsf{b}})$.

Semantics: Mixed Idiosyncratic Frames

Let W, Z be disjoint, nonempty sets and $R \subseteq (W \cup Z) \times Z$. $F = \langle W, Z, R \rangle$ is called a **mixed frame**. F is called a **mixed idiosyncratic frame** iff

(FUNCTIONALITY) $\forall w \in W \, \exists! v \in Z(wRv)$

(IDIOSYNCRACY) $\forall w, v \in Z(wRv \leftrightarrow w = v)$

Semantics: Mixed Idiosyncratic Frames

Let W, Z be disjoint, nonempty sets and $R \subseteq (W \cup Z) \times Z$. $F = \langle W, Z, R \rangle$ is called a **mixed frame**. F is called a **mixed idiosyncratic frame** iff

(FUNCTIONALITY) $\forall w \in W \, \exists! v \in Z (wRv)$

(IDIOSYNCRACY) $\forall w, v \in Z (wRv \leftrightarrow w = v)$

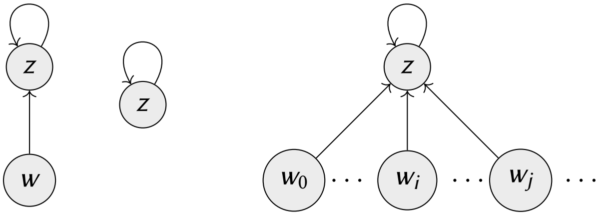

Figure: Idiosyncratic frame configurations

Mixed Models—FDE

- **Mixed Valuation**. $V : \mathrm{At} \times W \cup Z \to \{0, \mathrm{n}, \mathrm{b}, 1\}$ with $V_w(p) \in \{0, 1\}$ for all $p \in \mathrm{At}$ and $w \in W$.

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$\Box$ in classical worlds

Let $w \in W$. Then

$$\mathcal{M}, w \Vdash \Box\varphi \Leftrightarrow \forall v \in Z(wRv \Rightarrow \mathcal{M}, v \Vdash \varphi).$$

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$\Box$ in non-classical worlds

Let $z \in Z$. Then

$$\mathcal{M}, z \Vdash \Box\varphi \Leftrightarrow \forall v \in Z(zRv \Rightarrow \mathcal{M}, v \Vdash \varphi);$$
$$\mathcal{M}, z \Vdash \neg\Box\varphi \Leftrightarrow \exists v \in Z(zRv \,\&\, \mathcal{M}, v \Vdash \neg\varphi).$$

Truth in Mixed Models and Consequence—FDE

- $\mathcal{M} \Vdash_{\text{cl}} \varphi$ iff $\forall w \in W(\mathcal{M}, w \Vdash \varphi)$
- $\mathcal{M} \Vdash_{\text{fde}} \varphi$ iff $\forall z \in Z(\mathcal{M}, z \Vdash \varphi)$
- $F \Vdash^{+}_{\text{cl}} \varphi$ iff $\mathcal{M} \Vdash_{\text{cl}} \varphi$ for all $\mathcal{M}$ based on a **faithful valuation**.
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Consequence

Let Γ, Δ be finite sets of formulae. Then

- $\mathcal{M} \Vdash_{fde} (\Gamma \Rightarrow \Delta) :\Leftrightarrow \forall z \in Z\,[\forall \gamma \in \Gamma(\mathcal{M}, z \Vdash \gamma) \Rightarrow \exists \delta \in \Delta(\mathcal{M}, z \Vdash \delta)]$

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Remark on Variant Semantics

- K3: $V : \mathrm{At} \times W \cup Z \to \{0, \mathrm{n}, 1\}$;
- LP: $V : \mathrm{At} \times W \cup Z \to \{0, \mathrm{b}, 1\}$.

Completeness

Proposition

Let F be a mixed idiosyncratic frame. Then

$$F \Vdash_{\mathrm{fde}} \Gamma \Rightarrow \Delta \;\; \textit{iff} \;\; \mathsf{FDE}_{\Box} \vdash \Gamma \Rightarrow \Delta$$
$$F \Vdash^{+}_{\mathrm{cl}} \Gamma \Rightarrow \Delta \;\; \textit{iff} \;\; \mathsf{BM} \vdash \Gamma \Rightarrow \Delta.$$

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Corollary

Let F be a mixed idiosyncratic frame. Then

$$F \Vdash^{+}_{\mathrm{cl}} \varphi \Leftrightarrow \mathsf{BM} \vdash \varphi$$

Truth-theoretic Completeness

The Truth-theoretic Completeness of BM

Proposition

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- Soundness by induction the length of a proof in BM.
- Completeness via the contrapositive

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- $(\mathbb{N}, S^\dagger)$ is a model of KF, hence

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On Truth Tellers and the Realization $\circ$

Let $p_1, p_2, \ldots$ an enumeration of the propositional variables of $\mathcal{L}_\Box$.

$$p_j^\circ := \tau_{2j} \wedge \neg\tau_{2j+1}.$$

where τ_i given by

$$\mathsf{PAT} \vdash T\ulcorner\tau(\bar{i})\urcorner \leftrightarrow \tau(\bar{i}).$$

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Lemma

For any $X, Y \subseteq \omega$ there is a FDE*-fixed point S such that*

(*i*) $\#\tau_n \in S \Leftrightarrow n \in X$

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If $X \cap Y = \emptyset$ [$X \cup Y = \omega$], then we can find a consistent [complete] fixed point.

Finding the Fixed Point $S^{\dagger}$

Truth Teller Lemma

For any mixed, idiosyncratic, faithful model $\mathcal{M}$ of BM, we can find a FDE-fixed point $S \subset \omega$ such that:

$$
\begin{array}{ll}
(i) & \tau_{2j} \in S \Leftrightarrow \mathcal{M}, w \Vdash p_j \text{ or } \mathcal{M}, z \Vdash p_j; \\
(ii) & \neg\tau_{2j} \in S \Leftrightarrow \mathcal{M}, z \Vdash \neg p_j; \\
(iii) & \tau_{2j+1} \in S \Leftrightarrow \mathcal{M}, w \Vdash \neg p_j \ \& \ \mathcal{M}, z \Vdash p_j; \\
(iv) & \neg\tau_{2j+1} \in S \Leftrightarrow \mathcal{M}, z \Vdash p_j.
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S is called the image of $\mathcal{M}$.

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(iii)	$\tau_{2j+1} \in S \Leftrightarrow \mathcal{M}, w \Vdash \neg p_j \;\&\; \mathcal{M}, z \Vdash p_j$;
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- Choose $S^\dagger$ to be the image of $\mathcal{M}^\dagger$.

Connecting $\mathcal{M}^\dagger$ and $S^\dagger$

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Auxiliary Lemma

For all $\varphi \in \mathcal{L}_\Box$

(*i*) $\mathcal{M}^\dagger, z \Vdash \varphi \Rightarrow \#H^\circ(\varphi) \in S^\dagger$;

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Proof: Induction on the positive complexity of φ

- Faithful valuations
- Base cases by Truth Teller Lemma
- Modal cases by Auxiliary Lemma

The Strategy

- Assume $\mathrm{BM} \nvdash \varphi$.
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Corollary

Let Fix *be the set of Kripkean fixed points over* $\mathbb{N}$*. Then for all* $\varphi \in \mathcal{L}_\Box$

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- The same proof yields results for the pairs $(\mathrm{M^n}, \mathrm{KF} + \mathrm{CN})$ and $(\mathrm{M^b}, \mathrm{KF} + \mathrm{CM})$
- The proof can be lifted to the first-order case.

Refutation Proofs

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Consequences of Refutation Proofs

- M^{n} (M^{b}) is the modal logic of all consistent extensions of KF+CN (KF+CM).
- M^{n} is the modal logic of the sentences that are true in the minimal fixed-point model and the theory $\mathrm{KF}\mu$.

Refutation Proofs: Strategy

KF+CN

- Assume $\mathsf{M}^{\mathsf{n}} \nvdash \varphi$.
- By **completeness** there exists a mixed model $\mathcal{M}^{\ddagger}$ based on a faithful K3-valuation V such that for some $w \in W$

$$M^{\ddagger}, w \Vdash \neg\varphi$$

- Define a witness-realization $\ddagger$ depending on $\mathcal{M}^{\ddagger}$ such that

$$\mathsf{KF} + \mathsf{CN} \vdash \neg \mathsf{H}^{\ddagger}(\varphi)$$

- Hence

$$\exists * (\mathsf{KF} + \mathsf{CN} \nvdash \mathsf{H}^{*}(\varphi)).$$

The Witness-Realization

We assume faithful valuations. Let λ be a sentence such that

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KF+CN and the Liar

- $\mathsf{KF} + \mathsf{CN} \vdash \lambda$;
- $\mathsf{KF} + \mathsf{CN} \vdash \neg \mathrm{T}\ulcorner\lambda\urcorner$;
- $\mathsf{KF} + \mathsf{CN} \vdash \neg \mathrm{T}\ulcorner\neg\lambda\urcorner$.

Refutation Proofs: Lemmata

Auxiliary Lemma

For all $\varphi \in \mathcal{L}_\Box$

(*i*) $\mathcal{M}^{\ddagger}, z \Vdash \varphi \Rightarrow \mathsf{KF} + \mathsf{CN} \vdash \mathsf{T}\ulcorner H^{\circ}(\varphi)\urcorner;$

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Main Lemma

For all $\varphi \in \mathcal{L}_{\Box}$

$$\mathcal{M}^{\ddagger}, w \Vdash \varphi \Rightarrow \mathsf{KF} + \mathsf{CN} \vdash \mathrm{H}^{\circ}(\varphi).$$

- Truth-theoretic completeness of M^{n} without appealing to fixed-point models.
- Truth-theoretic completeness of M^{b} can be established by a similar argument.

The Logic of Grounded of Truth

Proposition (Nicolai & S.)

Let I *be the minimal fixed point of* $\mathcal{J}_{\mathrm{k3}}$*. Then for all* $\varphi \in \mathcal{L}_{\Box}$

$$\mathsf{M}^{\mathsf{n}} \vdash \varphi \Leftrightarrow \forall * ((\mathbb{N}, \mathsf{I}) \vDash H^{*}(\varphi));$$
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Proof of $\Leftarrow$.

Assume $\mathsf{M}^{\mathsf{n}} \nvdash \varphi$. Then $\mathsf{KF} + \mathsf{CN} \vdash \neg \mathsf{H}^{\ddagger}(\varphi)$. But $(\mathbb{N}, \mathsf{I})$ is a model of $\mathsf{KF} + \mathsf{CN}$ and thus $(\mathbb{N}, \mathsf{I}) \nvDash H^{\ddagger}(\varphi)$. *qed.*

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Proof of $\Leftarrow$.

Assume $\mathsf{M^n} \nvdash \varphi$. Then $\mathsf{KF} + \mathsf{CN} \vdash \neg \mathsf{H}^\ddagger(\varphi)$. But $(\mathbb{N}, \mathsf{I})$ is a model of $\mathsf{KF} + \mathsf{CN}$ and thus $(\mathbb{N}, \mathsf{I}) \not\models H^\ddagger(\varphi)$. *qed.*

- $\mathsf{M^b}$ is the logic of the maximal fixed point.
- $\mathsf{M^n}$ and $\mathsf{M^b}$ are a maximal truth logics:

The Logic of Grounded of Truth

Proposition (Nicolai & S.)

Let I *be the minimal fixed point of* $\mathcal{J}_{\mathrm{k3}}$*. Then for all* $\varphi \in \mathcal{L}_{\Box}$

$$\mathsf{M}^{\mathsf{n}} \vdash \varphi \Leftrightarrow \forall * ((\mathbb{N}, \mathsf{I}) \vDash H^{*}(\varphi));$$
$$\mathsf{M}^{\mathsf{n}} \vdash \varphi \Leftrightarrow \forall * (\mathsf{Kf}\mu \vdash H^{*}(\varphi)).$$

Proof of $\Leftarrow$.

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- M^{b} is the logic of the maximal fixed point.
- M^{n} and M^{b} are a maximal truth logics:
  - No proper extension of M^{n} or M^{b} can be the modal logic of a consistent truth theory.

Maximality or approximating Convention T

- Maximal truth logics **approximate Convention** T **as far as is consistently possible**;
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  - there is no consistent truth theory with a truth logic extending a maximal truth logic;
- Prima facie maximal truth logic are defensible weakenings of Convention T;
  - Uniqueness lost: rival conceptions of truth;
  - Philosophically motivated truth-maximal logics.

Maximality and Completeness beyond Kripkean Truth

Proposition (Czarnecki & Zdanowski; Standefer)

For all $\varphi \in \mathcal{L}_\Box$

$$\mathsf{KDD_c} \vdash \varphi \Leftrightarrow \forall * (\mathsf{FS} \vdash H^\star(\varphi));$$
$$\mathsf{KDD_c} \vdash \varphi \Leftrightarrow \forall * \exists n \in \omega \forall S \subseteq \omega ((\mathbb{N}, \mathcal{J}^n_{\mathrm{cl}}(S)) \models H^\star(\varphi));$$

and KDD_c *is a maximal truth logic.*

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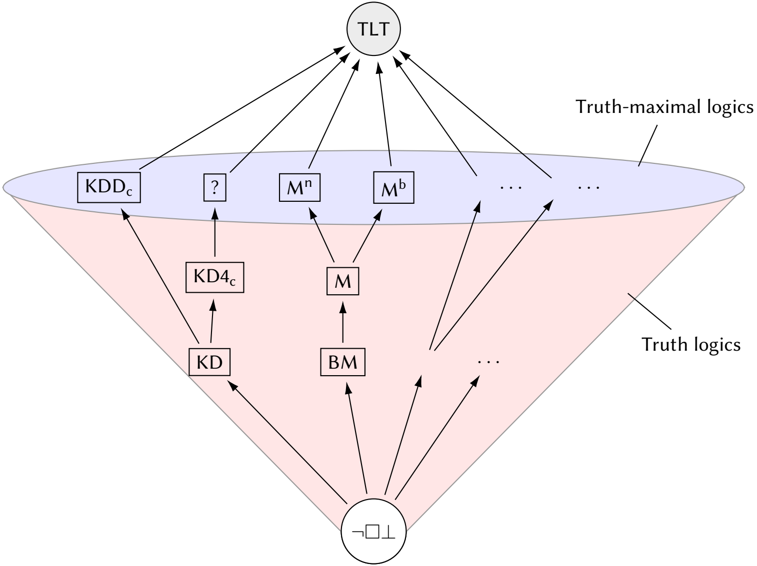
- $\mathsf{KDD_c}$ is the modal logic of FS and the finite stages of the Revision Theory of Truth;
- $\mathsf{KDD_c}$ is a maximal truth logic;
- A fully detailed rival account to Kripkean Truth

Logics of Truth



TLT is the modal logic axiomatized by $\Box\varphi \leftrightarrow \varphi$.

Conclusion

Summary

- Truth-theoretic completeness for the cluster of KF-theories
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Further results

- First-order logic of truth
- Weak Kleene and WKF
- Results for FWK and DT/FKF

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Canonical Model Construction and Existence Lemma

Canonical Model

$\mathcal{M}^C = (\langle W^C, Z^C, R^C \rangle, V^C)$ with

- W^C is the set of BM-maximal consistent sets.
- Z^C is the set of $\mathrm{FDE}_{\Box}$-saturated sets.
- For $v \in W^C \cup Z^C,\ z \in Z^C : vR^C z \Leftrightarrow \{\varphi : \Box\varphi \in v\} \subseteq z$
- $$V^C(v,p) := \begin{cases} 1, & \text{if } p \in v \,\&\, \neg p \notin v; \\ \mathrm{b}, & \text{if } p \in v \,\&\, \neg p \in v; \\ \mathrm{n} & \text{if } p \notin v \,\&\, \neg p \notin v; \\ 0, & \text{if } p \notin v \,\&\, \neg p \in v \end{cases}$$

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Existence Lemma

Let $w \in W^C$ and $\neg\Box\psi \in w$. Then there exists $z \in Z_C$ such that $wR^C z$ and $\psi \notin z$.

- Connecting Lemma!